**Superficially, selecting 1,000 models from such a large list of possibilities sounds very much like a data mining process.**

Data mining can be a very positive process. It is just that most people who do data mining are terrible at it. There are various things that you can do to make the data mining process work. It doesn't work on every data set. For some data sets, there is simply no edge available. Luckily for me, my intuition that there was an edge available in a nontrend following, noncountertrend type of model was correct. My intuition was that there should be other emergent patterns in the price data that are more complex than trend following.

**What are some of the things you can do to avoid the pitfalls of data mining?**

The first thing you need to do is to get some idea of how much of the apparent edge is completely spurious.

**How do you do that?**

Let's say instead of training with the target variable, which is the price change over the subsequent 24 hours, I generate random numbers that have the same distribution characteristics. I know that any models that I find that score well training on this data are 100 percent curve-fitted because they are based on intentionally bogus data. The performance of the best model on the fictitious data provides a baseline. Then, you need to come up with models that do much better than this baseline when you are training on the real data. It is only the performance *difference* between the models using real data and the baseline that is indicative of expected performance, not the full performance of the models in training.

**What are some of the worst errors people make in data mining?**

A lot of people think they are okay because they use in-sample data for training and out-of-sample data for testing.[6] Then they sort the models based on how they performed on the in-sample data and choose the best ones to test on the out-of-sample data. The human tendency is to take the models that continue to do well in the out-of-sample data and choose those models for trading. That type of process simply turns the out-of-sample data into part of the training data because it cherry-picks the